

8-A.1 $N = 10^9$, average degree $K = 200$

① Average path length $d = \frac{\log(N)}{\log(K)}$, $d = \frac{\log(10^9)}{\log(200)} = \frac{20.7}{5.3} = 3.9$ hops

② Milgram's = 6
 $N = 2 \cdot 10^8$

Which means the formula would give ≈ 36 hops.

③ Real networks are shorter when hubs acts as super-shortcuts.

——— || — longer when high clustering traps walks inside communities.

④ $d = \frac{\log(10^9)}{\log(10)} = 9$. More than double the path length

8-A.2 Code

Takeway: Pure ring lattice, Very ordered, high clustering, but long path lengths. So, not small-world

- Random graph, Very mixed, short path lengths, but low clustering. So, not small-world

- Karate-Club, short-ish path, noticeable clustering $\Rightarrow$ small world

Sign of Small world

Some clustering/communities but also has fast global reach

8-B.1 20-node ring lattice, 4 nearest neighbour connections each. Avg path = 5. Clustering = 0.5

① A single shortcut connecting maximally distant nodes (distance = $\frac{N}{2}$) can roughly halve the diameter. Because now there is a shortcut connecting distant nodes.

② Why clustering falls slowly. One rewired edge only destroys clusters touching said edge. So, path length (global) $\rightarrow$ sensitive to a long bridge while clusters are robust to a few rewires.

③ As rewiring probability (p) increases from 0 to 1 shortest path (L) drops fast early, while clustering stays high initially, then declines later. $\langle p \rangle$ approx $\langle c \rangle (1-p)^3$. So, at $p=0.01$ C is still about 0.97 $\langle c \rangle$ which means the clustering is still high meanwhile the L may have dropped a lot

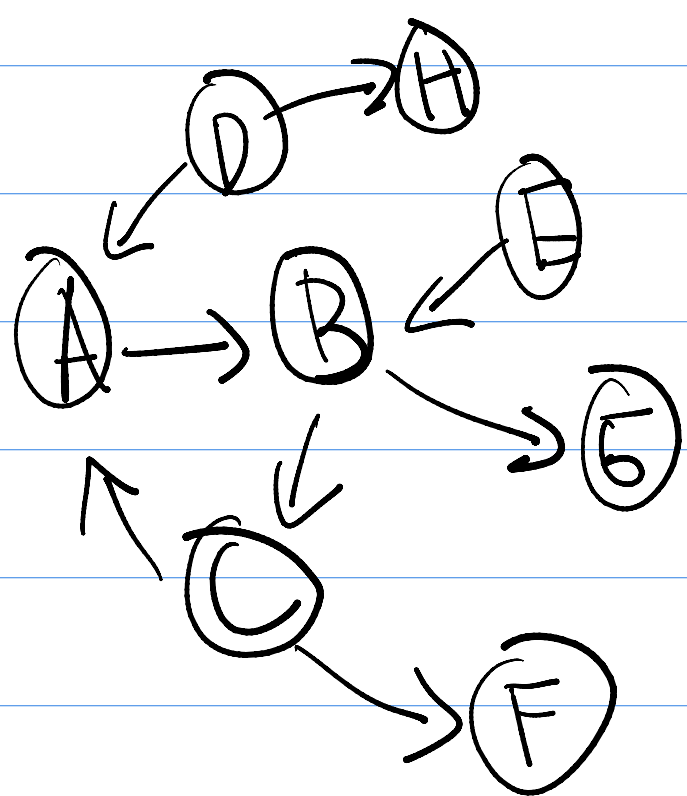
④ Real social network: small-world regime. high local clustering + rare long range weak

ties.

Q B2 < Code >

$L(p)$ collapses sharply while $C(p)$ stays high until much larger value of p .

Q-D-1 ①.



1.) Non-trivial SCC : $\{A, B, C\}$

2.) $\{A, B, C\} = \text{Main SCC}$

$\{D, E\} = \text{In}$

$\{F, G\} = \text{Out}$

$\{H\} = \text{Tendrils, only reachable from D}$

$\{I\} = \text{isolated}$

3.) $\{D, E\}$ can reach SCC

SCC reaches $\{F, G\}$

$\{D\}$ reaches $\{H\}$

4.) In = pages that links to hubs but receives no link back

Out = Pages that the SCC links but which doesn't link back

SCC = mutually reachable hub of major sites

Tendrils = Reachable subwebs that can only be reached from In

Q D2 < Codes >

In Component: pages that can reach SCC but cannot be reached from it, Pages from In are invisible to crawlers that start from the SCC.